Minimise:

$$[+1\alpha_1^2 + 2\alpha_1\alpha_2 - 5.5\alpha_1\alpha_3 - 6\alpha_1\alpha_4 + 2\alpha_2\alpha_4[?] + 4\alpha_2^2 - 11\alpha_2\alpha_3 - 12\alpha_2\alpha_4$$
$$_1 - 11\alpha_3\alpha_2 + 32.5\alpha_3^2 + 36\alpha_3\alpha_4 - 6\alpha_4\alpha_1 - 12\alpha_4\alpha_2 + 36\alpha_4\alpha_3 + 40\alpha_4^2] - [\alpha_1 + \alpha_2 + \alpha_3 + \alpha_4]$$

Subject to:

$\sum_i \alpha_i y_i = 0$

$$\alpha_1 + \alpha_2 - \alpha_3 - \alpha_4 = \upsilon$$

$\alpha_i \geq \upsilon$

$$\alpha_1 \leq 1,\ \alpha_2 \leq 1,\ \alpha_3 \leq 1,\ \alpha_4 \leq 1$$

$\alpha_i \leq \upsilon$

$$\alpha_1 > 0,\ \alpha_2 > 0,\ \alpha_3 > 0,\ \alpha_4 > 0$$

Quadratic Equation in α with constraints

$$\frac{1}{2}[X^T P X] + q^T X$$

Minimise:

$\frac{1}{2}$

$$[+2\alpha_1^2 + 4\alpha_1\alpha_2 - 11\alpha_1\alpha_3 - 12\alpha_1\alpha_4 + 4\alpha_1\alpha_4[?] + 8\alpha_2^2 - 22\alpha_2\alpha_3 - 24\alpha_2\alpha_4$$
$$_1 - 22\alpha_3\alpha_2 + 65\alpha_3^2 + 72\alpha_3\alpha_4 - 12\alpha_4\alpha_1 - 24\alpha_4\alpha_2 + 72\alpha_4\alpha_3 + 80\alpha_4^2] - [\alpha_1 + \alpha_2 + \alpha_3 + \alpha_4]$$

$$P = \begin{pmatrix} 2 & 4 & -11 & -12 \\ 4 & 8 & -22 & -24 \\ -11 & -22 & 65 & 72 \\ -12 & -24 & 72 & 80 \end{pmatrix}$$

$$q = [-1 \quad -1 \quad -1 \quad -1]$$